

## RUDNAYA PRISTAN, Russian Federation

WMO: 319590

Lat: 44.3690N

Lon: 135.8344E

Elev: 34

StdP: 100.92

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |       |      |       | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|-------|------|-------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |       | 1%   |       |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB  | WS   | MCDB  | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)   | (l)  | (m)   | (n)                   | (o)  | (p)   |
| 1             | -19.9      | -18.0 | -30.3                         | 0.2 | -17.5 | -28.6 | 0.3 | -16.2 | 11.5                  | -14.6 | 10.6 | -14.3 | 4.3                   | 270  | 0.554 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 6.0                    | 26.5            | 19.7 | 24.3 | 19.1 | 22.6 | 18.4 | 21.4                | 24.0 | 20.4 | 22.9 | 19.5 | 21.7 | 2.4                  | 270  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 20.5                            | 15.3 | 22.5 | 19.5 | 14.3 | 21.5 | 18.7 | 13.6 | 20.6 | 62.5          | 24.2 | 59.1 | 23.0 | 56.1 | 21.7 | 25.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |     |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-----|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |     |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB  | WB  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |     |     |
| (4) 8.9           | 7.5  | 6.4 | -23.6                      | 30.9 | 2.4                | 2.0 | -25.3                                              | 32.4 | -26.7      | 33.5 | -28.0      | 34.6 | -29.8      | 36.1 |     |     |
| (5)               |      |     | -24.0                      | 23.1 | 2.3                | 1.2 | -25.6                                              | 24.0 | -27.0      | 24.7 | -28.2      | 25.4 | -29.9      | 26.2 | (4) | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 4.6    | -10.4 | -7.5  | -1.9  | 4.0   | 8.3   | 12.2  | 16.9  | 19.1  | 15.1  | 7.9   | -1.1  | -8.5  | (6)  |  |
|                                                                        | DBStd       | 10.51  | 3.94  | 4.00  | 3.67  | 2.95  | 3.34  | 2.91  | 2.70  | 2.54  | 3.01  | 3.54  | 4.98  | 4.29  | (7)  |  |
|                                                                        | HDD10.0     | 2755   | 632   | 491   | 369   | 183   | 76    | 9     | 0     | 0     | 2     | 85    | 334   | 574   | (8)  |  |
|                                                                        | HDD18.3     | 5099   | 890   | 724   | 628   | 431   | 312   | 188   | 62    | 20    | 103   | 324   | 584   | 833   | (9)  |  |
|                                                                        | CDD10.0     | 767    | 0     | 0     | 0     | 2     | 22    | 73    | 212   | 282   | 156   | 19    | 0     | 0     | (10) |  |
|                                                                        | CDD18.3     | 70     | 0     | 0     | 0     | 0     | 1     | 2     | 16    | 44    | 7     | 0     | 0     | 0     | (11) |  |
|                                                                        | CDH23.3     | 294    | 0     | 0     | 0     | 0     | 9     | 24    | 76    | 156   | 28    | 1     | 0     | 0     | (12) |  |
|                                                                        | CDH26.7     | 56     | 0     | 0     | 0     | 0     | 2     | 7     | 16    | 27    | 4     | 0     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 2.7    | 4.1   | 3.3   | 2.9   | 2.4   | 2.3   | 1.9   | 1.7   | 1.8   | 2.2   | 2.6   | 3.1   | 3.6   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 768    | 17    | 14    | 27    | 50    | 84    | 91    | 125   | 119   | 110   | 68    | 39    | 23    | (15) |  |
|                                                                        | PrecMax     | 1172   | 79    | 48    | 88    | 150   | 180   | 324   | 529   | 219   | 296   | 272   | 120   | 64    | (16) |  |
|                                                                        | PrecMin     | 426    | 0     | 0     | 0     | 3     | 2     | 27    | 30    | 43    | 10    | 5     | 3     | 0     | (17) |  |
|                                                                        | PrecStd     | 168    | 17    | 14    | 22    | 35    | 43    | 65    | 91    | 48    | 81    | 52    | 29    | 19    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 1.2   | 5.9   | 11.6  | 18.2  | 24.0  | 26.7  | 28.9  | 29.5  | 26.5  | 20.2  | 13.2  | 4.8   | (19) |  |
|                                                                        |             | MCWB   | -2.3  | 0.7   | 4.4   | 8.8   | 12.6  | 17.4  | 20.9  | 20.9  | 18.2  | 12.5  | 7.0   | 1.6   | (20) |  |
|                                                                        | 2%          | DB     | -0.6  | 2.6   | 7.8   | 14.0  | 19.7  | 21.4  | 25.4  | 26.9  | 23.6  | 17.3  | 10.5  | 2.0   | (21) |  |
|                                                                        |             | MCWB   | -3.0  | -1.4  | 2.3   | 6.7   | 11.3  | 15.8  | 19.7  | 20.6  | 16.9  | 11.5  | 6.1   | -0.9  | (22) |  |
|                                                                        | 5%          | DB     | -2.2  | 0.6   | 5.5   | 11.4  | 16.5  | 18.2  | 22.6  | 24.9  | 21.8  | 15.6  | 8.6   | 0.3   | (23) |  |
|                                                                        |             | MCWB   | -4.6  | -2.3  | 0.9   | 5.4   | 9.9   | 14.5  | 18.9  | 20.0  | 16.1  | 10.8  | 5.1   | -2.2  | (24) |  |
|                                                                        | 10%         | DB     | -3.9  | -1.0  | 3.5   | 9.2   | 13.6  | 16.1  | 20.5  | 23.2  | 20.3  | 14.1  | 6.7   | -1.6  | (25) |  |
|                                                                        |             | MCWB   | -6.1  | -3.4  | -0.2  | 4.2   | 8.9   | 13.6  | 18.3  | 19.6  | 15.7  | 10.0  | 3.8   | -4.0  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | -1.0  | 1.0   | 5.1   | 9.6   | 14.2  | 18.6  | 21.8  | 23.1  | 20.5  | 14.6  | 9.0   | 2.2   | (27) |  |
|                                                                        |             | MCDB   | 0.0   | 4.2   | 10.4  | 16.4  | 22.5  | 24.2  | 26.3  | 26.4  | 23.5  | 17.2  | 11.0  | 4.0   | (28) |  |
|                                                                        | 2%          | WB     | -2.8  | -0.7  | 2.8   | 7.2   | 11.9  | 16.6  | 20.3  | 21.9  | 18.9  | 13.2  | 7.0   | -0.4  | (29) |  |
|                                                                        |             | MCDB   | -0.8  | 1.6   | 7.1   | 12.6  | 17.3  | 20.3  | 23.6  | 24.2  | 21.3  | 15.9  | 9.5   | 1.6   | (30) |  |
|                                                                        | 5%          | WB     | -4.6  | -1.9  | 1.4   | 6.0   | 10.6  | 15.1  | 19.4  | 21.1  | 17.9  | 12.0  | 5.5   | -2.2  | (31) |  |
|                                                                        |             | MCDB   | -2.4  | 0.4   | 4.3   | 10.2  | 14.8  | 17.6  | 21.6  | 23.4  | 20.0  | 14.3  | 8.3   | 0.3   | (32) |  |
|                                                                        | 10%         | WB     | -6.1  | -3.4  | 0.5   | 4.9   | 9.6   | 13.8  | 18.5  | 20.3  | 17.0  | 10.7  | 3.7   | -3.9  | (33) |  |
|                                                                        |             | MCDB   | -3.9  | -0.9  | 2.8   | 8.0   | 12.8  | 15.7  | 20.3  | 22.4  | 19.3  | 13.2  | 6.7   | -1.6  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 8.4   | 9.5   | 8.6   | 8.1   | 6.9   | 5.4   | 4.8   | 6.0   | 9.4   | 10.1  | 9.1   | 8.4   | (35) |  |
|                                                                        |             | MCDBR  | 10.6  | 11.2  | 11.6  | 12.6  | 13.1  | 10.7  | 9.7   | 9.3   | 11.3  | 12.3  | 11.7  | 10.7  | (36) |  |
|                                                                        |             | MCWBR  | 8.9   | 8.8   | 7.5   | 7.0   | 6.1   | 5.8   | 4.8   | 4.5   | 5.9   | 7.8   | 8.5   | 8.8   | (37) |  |
|                                                                        | 5% WB       | MCDBR  | 10.6  | 11.0  | 10.3  | 11.2  | 10.8  | 9.0   | 7.6   | 7.1   | 8.2   | 10.4  | 11.1  | 10.7  | (38) |  |
|                                                                        |             | MCWBR  | 9.2   | 9.1   | 7.2   | 6.7   | 5.7   | 5.5   | 4.3   | 4.1   | 5.4   | 7.8   | 8.7   | 9.1   | (39) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.310 | 0.372 | 0.445 | 0.539 | 0.566 | 0.542 | 0.503 | 0.465 | 0.413 | 0.418 | 0.370 | 0.328 | (40) |  |
|                                                                        | taud        |        | 2.224 | 2.037 | 1.895 | 1.757 | 1.776 | 1.924 | 2.106 | 2.219 | 2.281 | 2.104 | 2.147 | 2.155 | (41) |  |
|                                                                        | Ebn at Noon |        | 789   | 784   | 771   | 730   | 727   | 748   | 775   | 792   | 805   | 725   | 697   | 712   | (42) |  |
|                                                                        | Edh at Noon |        | 88    | 129   | 170   | 213   | 215   | 187   | 154   | 133   | 115   | 120   | 96    | 86    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 2.10  | 3.09  | 4.13  | 4.62  | 4.88  | 4.92  | 4.55  | 4.47  | 4.12  | 3.06  | 2.10  | 1.70  | (44) |  |
|                                                                        | RadStd      |        | 0.13  | 0.19  | 0.29  | 0.45  | 0.48  | 0.67  | 0.51  | 0.37  | 0.36  | 0.25  | 0.18  | 0.11  | (45) |  |

## Historical Trends

|                             |     | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|-----|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |     |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (d)                         | (e) | (f)   | (g)     | (h)    | (i)     | (j)   | (k)   | (l)         | (m)     |         |         |
| (46) Station Only           | N/A | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (3 neighbors) | N/A | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page